

Discrete-time LQR state feedback control with integral action for asymptotic step tracking

1 Introduction

1. Definizione specifiche

```
Ts = 0.01e-3;
```

2. Definizione P nello state space

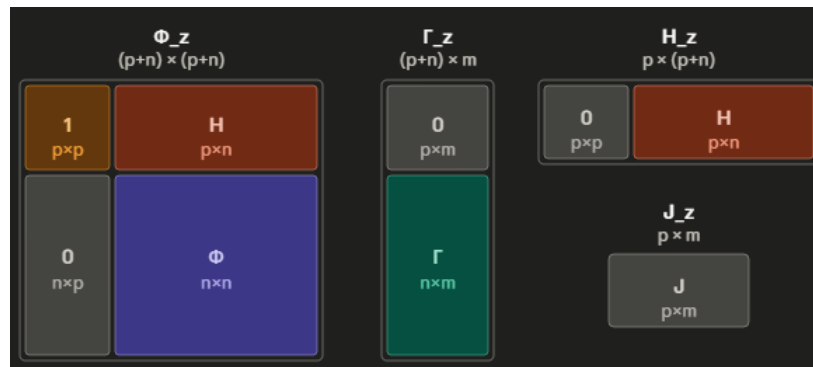
```
P_ss_c = ss(A,B,C,D);
```

```
P_ss_d = c2d(P_ss_c,Ts,'zoh');
```

```
[Phi, Gamma, H, J] = ssdata(P_ss_d);
```

```
n = size(Phi, 1); % ordine sistema
p = size(H, 1); % numero uscite
m = size(Gamma, 2); % numero ingressi
```

3. Costruzione matrici estese



```
Phi_z = [1, H; zeros(n,1), Phi];
Gamma_z = [zeros(p,m);Gamma];
H_z = [zeros(p,p),H];
J_z = J;
```

4. Definizione Q ed R

$$Q = \underbrace{\begin{bmatrix} \frac{1}{10} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{10} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{10} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{10} \end{bmatrix}}_{n+1} \left. \vphantom{\begin{bmatrix} \frac{1}{10} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{10} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{10} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{10} \end{bmatrix}} \right\} n+1$$

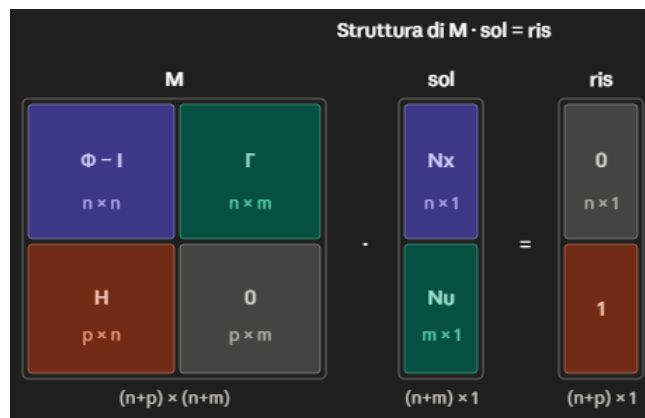
```
Q = diag(ones(1, n+1) / 10);
R = 1/20;
```

5. Calcolo feedback gain e poli catena chiusa

$$\mathbf{K}_z = \left[k_i \mid \underbrace{k_0 \quad k_1 \quad \cdots \quad k_{m-1}}_{\mathbf{K}} \right]$$

```
% Feedback gain and resulting closed loop poles
[Kz,~,poles_T_d] = dlqr(Phi_z,Gamma_z,Q,R);
Ki = Kz(1);
K = Kz(2:end);
```

6. Calcolo di Nx e Nu



```
% Nx, Nu
M = [Phi - eye(n), Gamma; H, J];

ris = [zeros(n, 1); 1];

sol = inv(M) * ris;

Nx = sol(1:n);
Nu = sol(n+1:end);
```

7. Full-state observer

```
% eigenvalues in z domain, 4-5 times faster than CL discrete poles
poles_d_obs = [0.2,0.25];

% Observer Gain
L = place(Phi',H',poles_d_obs)';

Phi_obs = Phi-L*H;
Gamma_obs = [Gamma, L];
H_obs = eye(n);
J_obs = zeros(n, m+p);
```

8. Simulink model

